

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (ME) Examination

September 2018

ME 310 Production Technology

Date: 25.09.2018, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following questions.

- (a) Enlist any four essential characteristics of cutting tool material. [02]
- (b) State the essential functions of cutting fluid used during machining. [02]
- (c) What is the importance of providing a clearance angle to single point cutting tool? [01]

Q – 2 Answer any four. [16]

- (a) Sketch a single point cutting tool and show on it the various tool elements and tool angles.
- (b) Write short note on “Types of Chips in metal cutting”.
- (c) Explain: how oblique cutting differs from orthogonal cutting?
- (d) Define a tool life. A tool life of 80 min is obtained at a speed of 30 m/min and 8 min at 60 m/min. Determine the tool life equation.
- (e) Explain progressively processing type of semi automat with neat sketch.
- (f) Enlist the selection criteria for materials for machine tool structure.

Q – 3 Answer any two. [14]

- (a) Derive an equation $\tan \phi = r \cos \alpha / (1 - r \sin \alpha)$ relating shear angle with chip thickness ratio and rake angle.
- (b) With usual notations draw and explain ‘Merchant’s circle diagram’.
- (c) Calculate the rpm values and diameter range served by each rpm by using arithmetic progression for the following conditions: $n_1 = 30$ rpm, $n_z = 375$ rpm, number of speed steps $z = 12$, $v = 20$ m/min.

SECTION – II

- Q – 4 (a)** Classify various modern machining processes. [03]
(b) Explain the axial gear hobbing process with neat sketch. [04]

- Q - 5 (a)** Describe setup and working of Electron Beam Machining (EBM) process with the help of sketch. [07]
(b) List the different types of jigs and explain any three drilling jigs. [07]

OR

- Q - 5 (a)** Draw general setup for an Electrical Discharge Machining (EDM) process and explain basic principle of EDM. [07]
(b) Explain the 3:2:1 principle of location with neat sketch. [07]

- Q - 6 (a)** List different process parameters of ultrasonic machining process and explain their effects on metal removal rate. [07]
(b) Describe thread rolling process and list its advantages and limitation. [07]

OR

- Q - 6 (a)** Explain the principle of plasma arc machining stating its applications, advantages and limitations. [07]
(b) Which criteria should be considered to design lathe fixture? Briefly explain lathe fixture with neat sketch. [07]
